Online Marketplaces

Part 5: Marketplace Design: Asymmetric Information & Reputation Systems

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This Lecture: Mechanism Design and Digital Advertising

2. **Information Asymmetry and Reputation Systems in Platforms**
   - Moral Hazard,
   - Adverse Selection
   - Signaling

Marketplace Design Background: Asymmetric Information

- Information asymmetry deals with the study of decisions in transactions where one party has more or better information than the other.
- When the information is asymmetrically distributed among the agents, the designing of contracts or allocations are intended to provide incentives and/or to induce the revelation of private information.
- Information asymmetry is typically divided into three main themes: moral hazard, signaling, and adverse selection.

Moral Hazard

- A moral hazard problem exists when the agent's action is not verifiable, therefore it cannot be included in the terms of the contract.
- Labor market: the effort of a worker (agent) is not verifiable (should wages be fixed?)

  Insurance: the insurance company wants the policyholder to try to avoid accidents. However, once a person is insured he has an incentive to be less cautious than before.

Moral Hazard in Online Platforms

- **Ride-Sharing Example (Uber, Lyft):**
  - Drivers are paid for rides, but the platform can't perfectly observe whether the driver is giving good service.
  - *Mechanism Design Issue:* Need monitoring!
- **Labor Platforms (Upwork, Fiverr):**
  - Hard for the hiring party to monitor effort.
  - The platform might introduce milestones or time-tracking software to mitigate moral hazard.
- Other examples?

How platforms mitigate this information friction?

Upwork Time Tracker

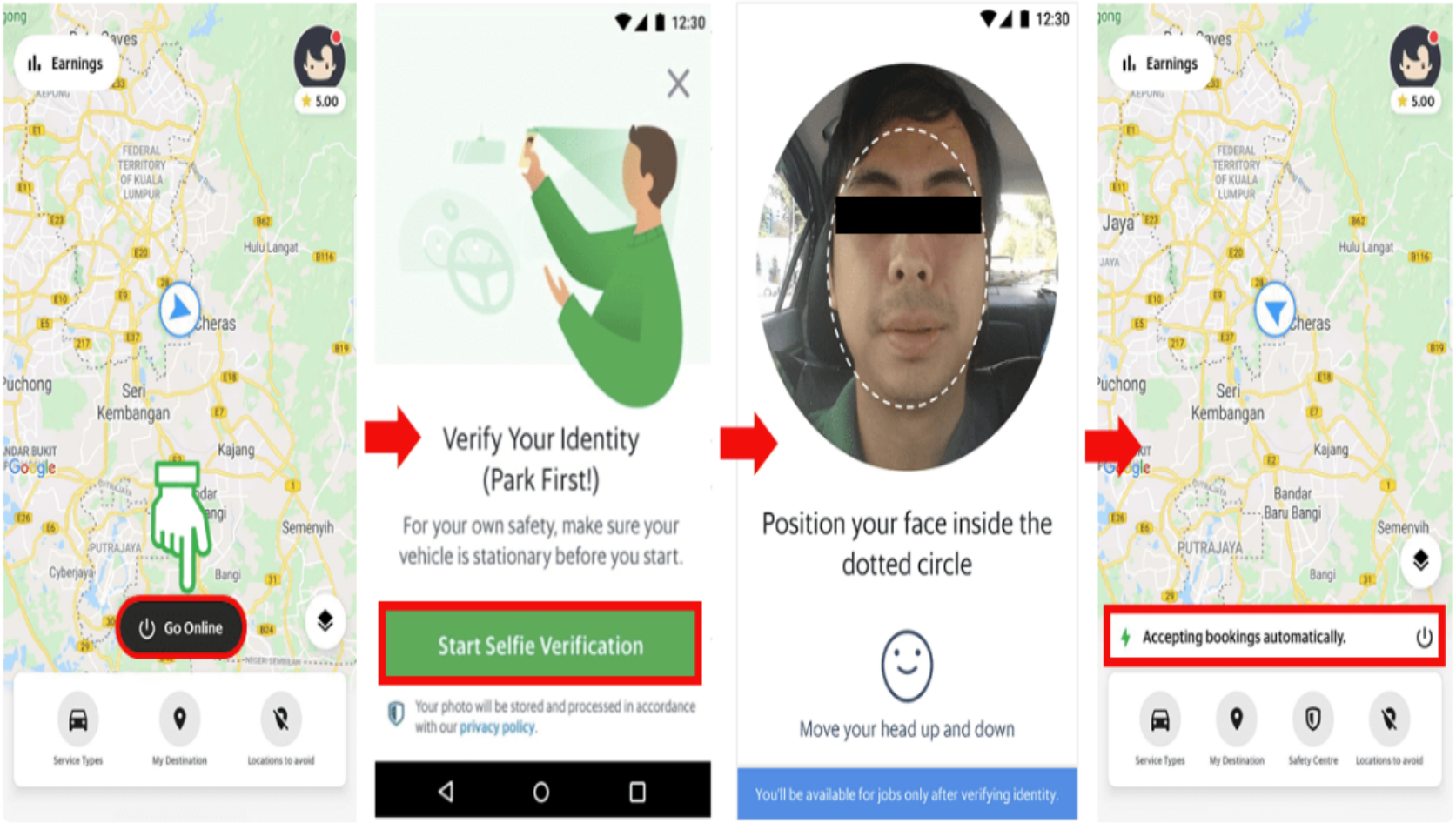
Review your freelancer's Work Diary

To help you monitor your freelancers' activity and smooth out the billing process, we offer the Work Diary. The Work Diary is a visual record of your freelancer's activity on a given project.

Here's how it works:

- You set up an hourly contract with your freelancer
- We request that your freelancer track their time via our Time Tracker
- We take real screenshots of what your freelancer is working on about six times an hour when your freelancer uses the Time Tracker. You can view these work-in-progress snapshots when you review your freelancer's Work Diary
- Your freelancer adds more details in their Work Diary about what they were working on

Grab's Driver Selfie Authentication Feature



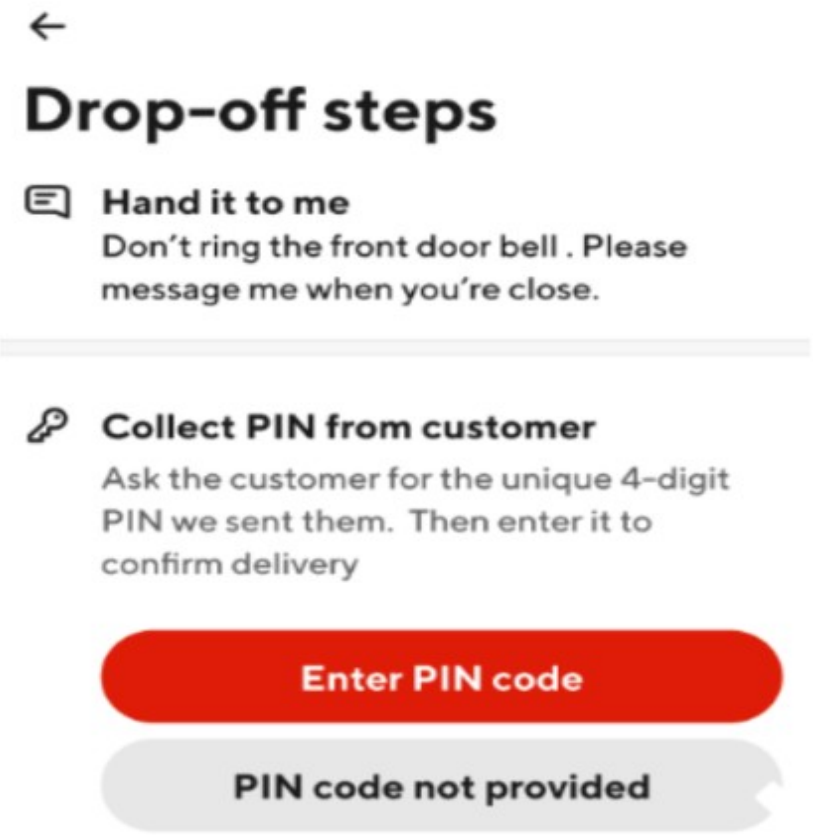
Rover Cards

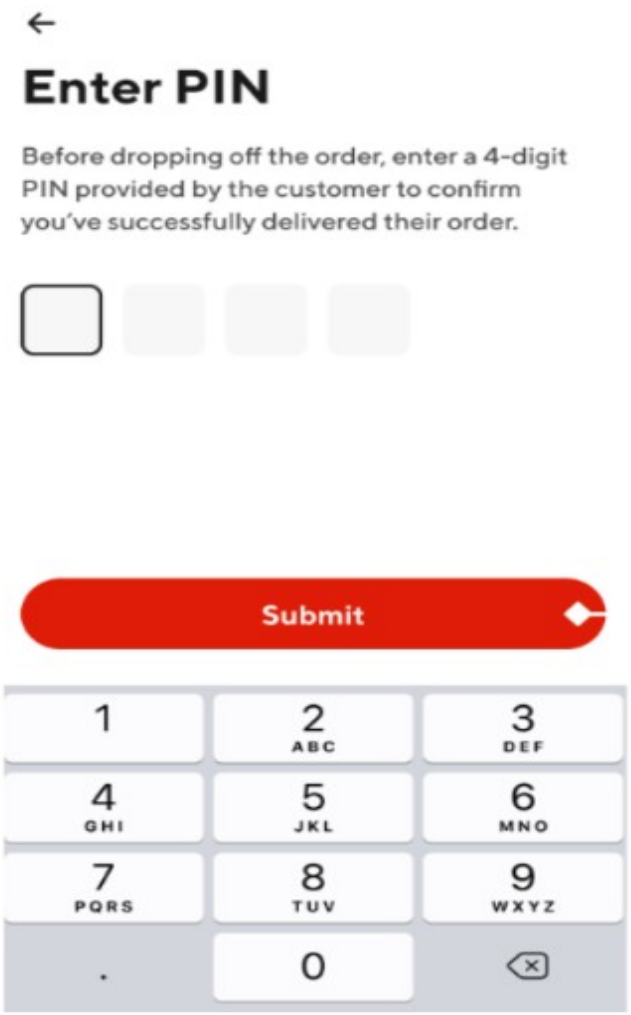
The dog walking map works with the Rover Card, a feature that allows you to share your walking route with pet owners. You can also share information about pee, poo, food, and water activity during the walk.

The map includes:

- The route
- Total time
- Distance of walk

DoorDash: From Soft Monitoring to "hard proof"





Adverse Selection

- Adverse selection occurs when one party in a transaction has more information about a key characteristic than the other.
- A lawyer is usually better informed than his client with respect to his personal characteristics.
- Insurance: A driver knows more than his insurance company about his driving habits.

  Regulation: A regulated monopoly has private information, the government doesn't have information about the monopoly's cost and demand.
- E-commerece: Sellers of used goods know more about the condition of their product than buyers do.

  How platforms mitigate this problem?

Adverse Selection in Online Marketplaces

Problem: Buyers face uncertainty about seller quality due to asymmetric information.

- Sellers have private information about product/service quality.
- High-quality sellers may be indistinguishable from low-quality sellers.

Feedback mechanisms reduce information asymmetry by:

- Aggregating past data. Buyers can make informed decisions based on seller history.
- Creating incentives for sellers to maintain high-quality service.

One-Sided vs. Two-Sided Feedback

Different platforms design feedback mechanisms based on transaction risks:

- **One-sided feedback (eBay, Amazon)** - Only buyers rate sellers.
- **Two-sided feedback (Airbnb, Uber)** - Both parties rate each other.

Trade-offs?

- **One-sided feedback:**
  - Simpler system but does not account for buyer behavior.
  - More prone to selection bias, only extreme experiences (very good or very bad) may be reviewed.
- **Two-sided feedback:**
  - Potential for "retaliatory ratings": users may fear giving negative feedback.
  - Platforms like Airbnb mitigate this by only revealing ratings after both parties submit feedback (or after x days).

eBay Moved to One-Sided Feedback

Before 2008: eBay used a two-sided feedback system where both buyers and sellers could rate each other.

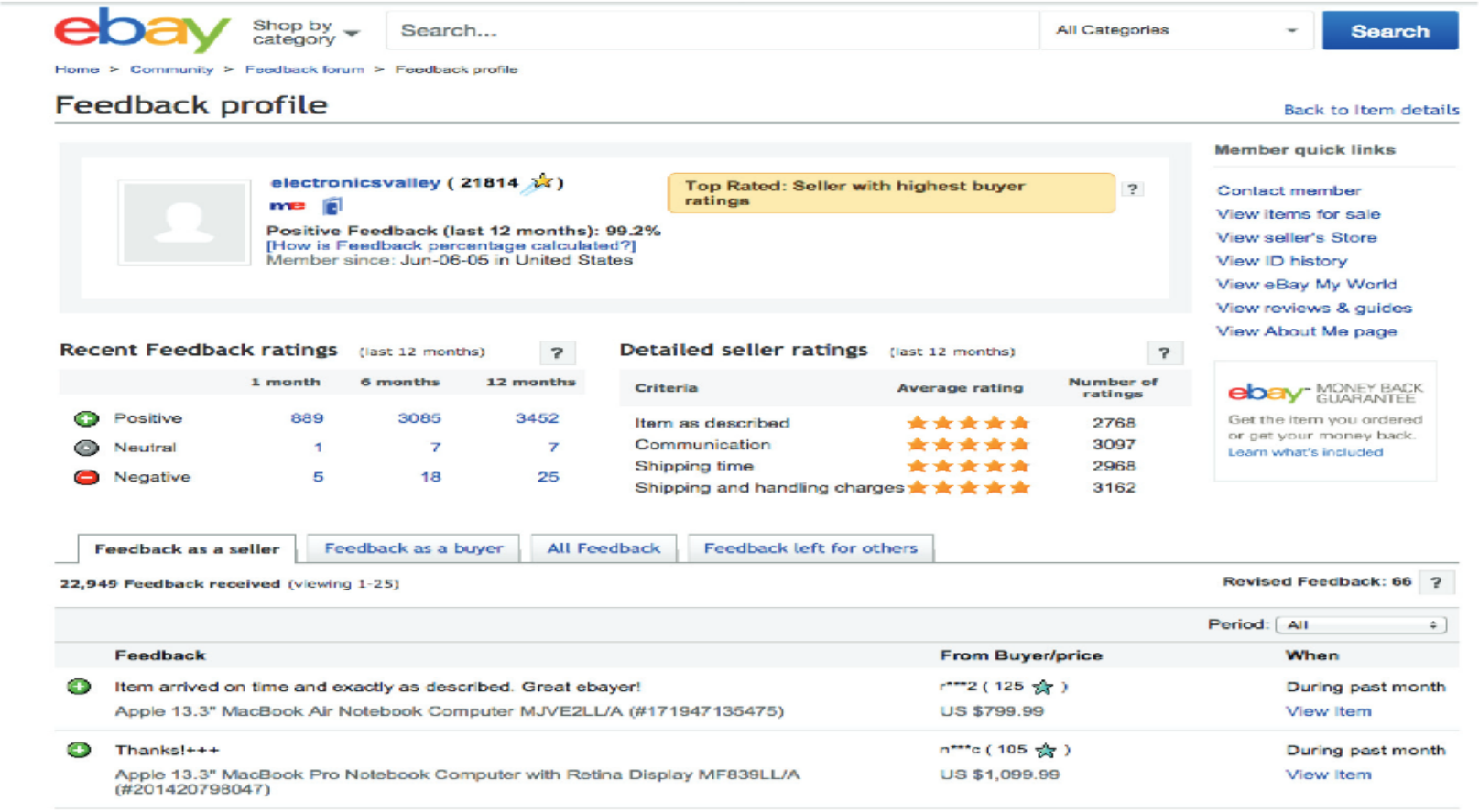
- Sellers could leave retaliatory negative feedback for buyers who left poor reviews.
- This discouraged buyers from leaving honest negative feedback.
- Created a bias where most reviews were overwhelmingly positive.

After 2008: eBay switched to a one-sided feedback system where only buyers could leave ratings.

- Reduced fear of retaliation, leading to more accurate feedback.
- Introduced Detailed Seller Ratings (DSRs).

eBay Seller Feedback Profile

- Reputation is summarized as **positive, neutral, or negative feedback**.
- **Detailed Seller Ratings (DSRs):** Anonymous ratings on multiple attributes. Publicly visible.



The Role of "No Reviews" as a Signal

Nosko and Tadelis (2016) found that silent transactions are often negative signals in eBay:

- Many dissatisfied buyers prefer to leave no feedback rather than a negative review.
- Standard reputation metrics that only count explicit ratings may overstate seller quality.

Effective Percent Positive (EPP) Metric:

- eBay introduced the EPP metric, which penalizes sellers for transactions with no feedback.
- $\text{EPP} = \frac{\text{Positive Feedback}}{\text{Total Transactions (Including No Reviews)}}$.

Comparing PP and EPP

Example: Two Sellers with the Same PP but Different EPP

- Seller A: 120 transactions, 99 positive, 1 negative, 20 silent transactions.
- Seller B: 150 transactions, 99 positive, 1 negative, 50 silent transactions.

Traditional Percent Positive (PP) Score:

- Both sellers have PP of $\frac{99}{99+1} = 99\%$, making them look equally reliable.

Effective Percent Positive (EPP) Score:

- Seller A: $\frac{99}{120} = 82.5\%$
- Seller B: $\frac{99}{150} = 66\%$
- Seller B appears worse in EPP, even though PP suggests they are equal.

PP vs. EPP in eBay

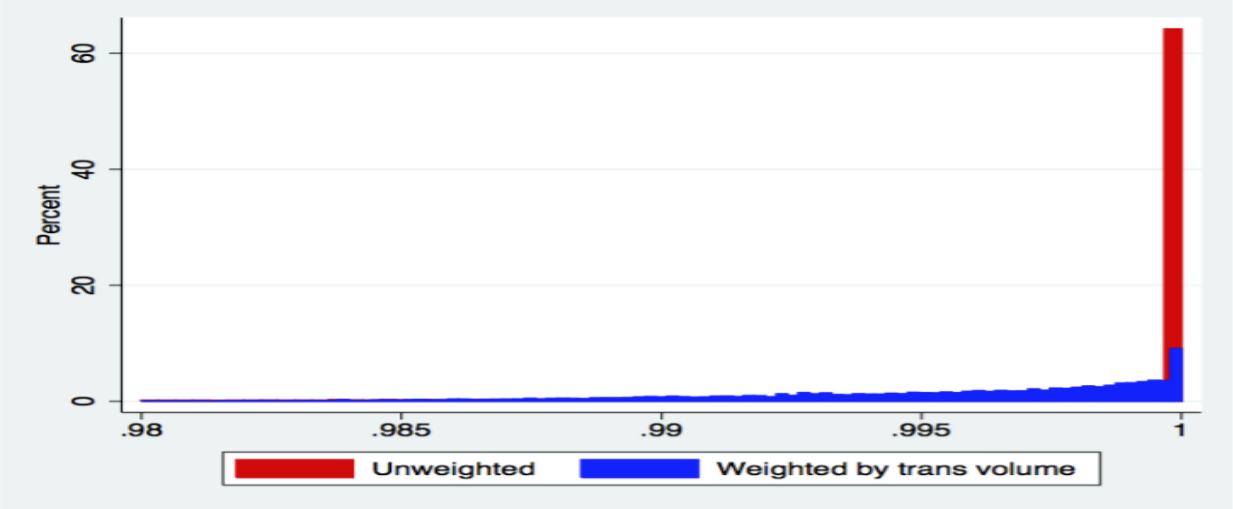


Figure 3: Percent Positive of Sellers

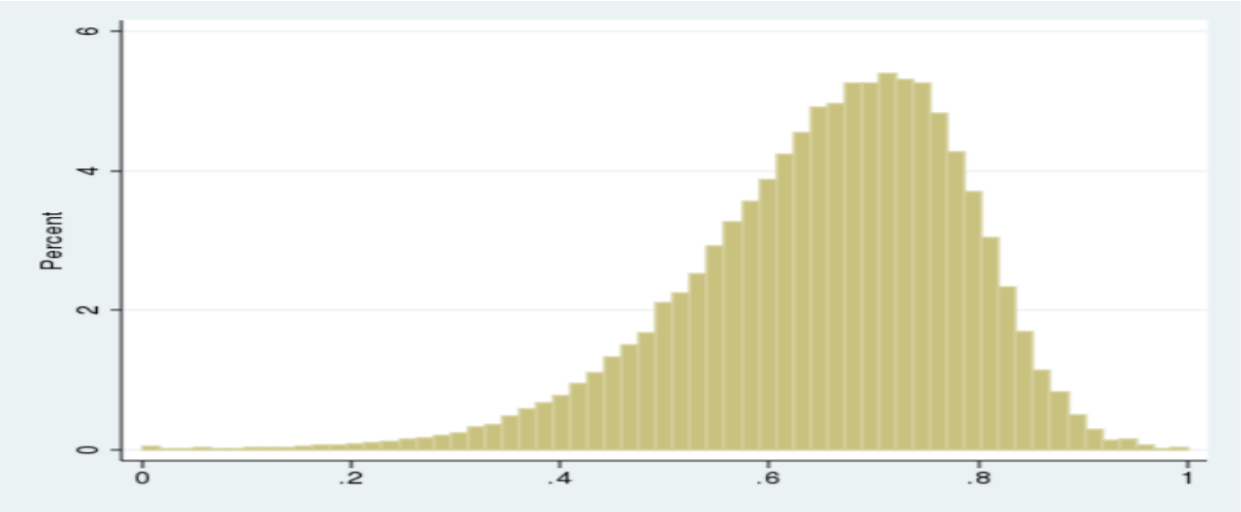


Figure 5: Histogram of Sellers' Effective Percent Positive Scores

EPP's Impact on eBay's Marketplace

How eBay Used EPP Internally:

- Search Ranking Algorithm: eBay used EPP to prioritize better sellers in search results.
- Improved Buyer Experience: Buyers exposed to high-EPP sellers were more likely to return.
- Encouraging Reviews: Buyers were incentivized to leave feedback (discounts, coupons, emails, etc).

EPP Was Not Publicly Available:

- eBay does not display total transactions, preventing buyers from calculating EPP themselves. Instead, eBay continued displaying the PP metric.
- Why EPP was not publicly available?

Upwork's Job Success Score (JSS)

- **Key mechanism: Private feedback at contract close**
  - Clients leave **public feedback** (stars/comments) that appears on the freelancer's profile.
  - Clients also leave **private feedback** that is **not shared**.
- **Internal score:** Upwork aggregates contract outcomes into a proprietary **Job Success Score (0–100%)**, using **public + private feedback** (plus other features like disputes, long-term relationships), and updates it daily.
- **How Upwork uses this:**
  - **Search & matching:** JSS is shown in search and the algorithm favors higher JSS in ranking/visibility.
  - **Badges / gating:** Eligibility for **Top Rated / Top Rated Plus** requires maintaining high JSS.
  - **Enforcement:** Repeated negative end-of-contract feedback can trigger a ban from Upwork.

Challenges in Reputation Systems

- **Feedback Bias:** Buyers may avoid leaving negative reviews due to retaliation.
  Possible solutions: One-sided review, anonymous feedback mechanisms (e.g., eBay's DSRs).
- **Fake Reviews:** Sellers may manipulate ratings through fake transactions.
  Possible solution: Using machine learning to detect fake reviews and rating manipulations.
- **Grade Inflation:** Overwhelmingly positive ratings reduce signal strength.
  Possible solution: experiment with internal metrics (e.g., eBay's EPP).
- **Information Overload:** Buyers may struggle to interpret complex rating data.
  Possible Solution: Simple visible metrics.

Signaling

- A signal is an activity or decision that credibly conveys "private" information about an agent or product's quality, ability, or characteristic to others.
- Labor: Firms search for workers with the ability to learn, and in order to signal this ability, some potential workers invest a few years in studying, even if the courses taken are of no use at all to the firms.

  Product Question: How can a consumer be induced to buy a product and learn that it is high quality?

  Amazon's "Vine Voice" or Verified Purchases and Amazon's Choice.

  Examples?

Buying Reputation as a Signal of Quality: Evidence from an Online Marketplace

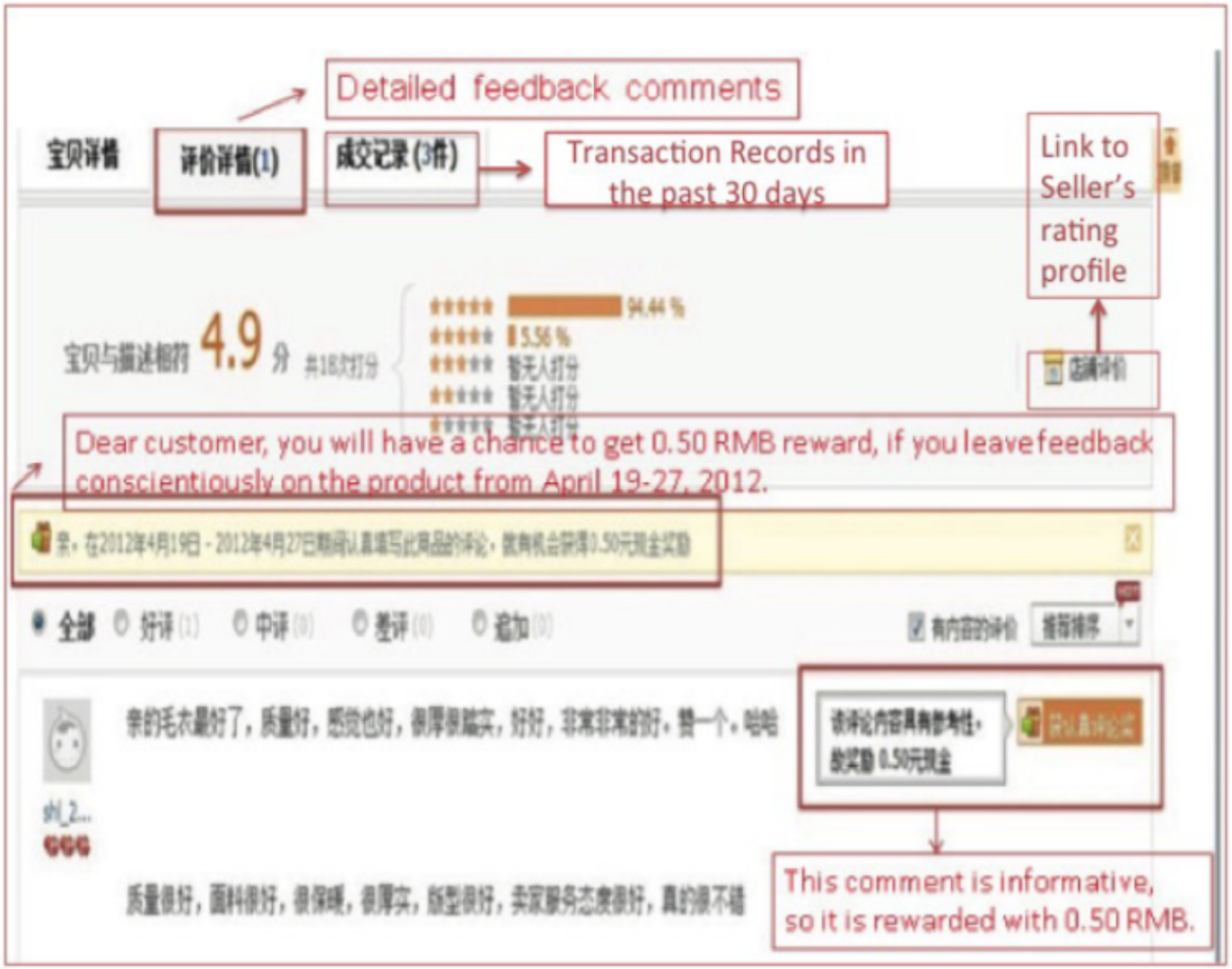
The Taobao Reward-for-Feedback (RFF) Mechanism introduced in March 2012 on Taobao (Alibaba's C2C platform).

Sellers offer rewards (cash or coupons) to buyers for leaving informative feedback.

Taobao ensures reward transfer based on feedback quality (measured by ML algorithm).

Hypothesis: Only high-quality sellers will use RFF since they expect positive reviews. Why?
Signaling theory implies that only high quality would signal..

Taobao RFF


Detailed feedback comments
宝贝详情
评价详情(1)
成交记录(3件)
Transaction Records in the past 30 days
Link to Seller's rating profile
宝贝与描述相符 4.9 分
Dear customer, you will have a chance to get 0.50 RMB reward, if you leave feedback conscientiously on the product from April 19-27, 2012.
全部
好评
中评
差评
追加
This comment is informative, so it is rewarded with 0.50 RMB.

Empirical Findings

Dataset: 6.99M transactions from 12,857 sellers in four product categories.

- Higher-quality sellers are more likely to offer rewards (RFF as a signaling tool).
- Items with RFF experience a **30% increase in sales**.
- RFF does not bias feedback positivity but increases the length and informativeness of reviews.
- Increase in **zero-character ratings** by 14.24% when RFF is offered (despite that RFF rewards only informative reviews).

Takeaway:

- High-quality sellers consistently offer RFF, expecting good feedback.
- Buyers may interpret RFF as a **credible signal of seller confidence in their product**.

Market Design Implications

- Marketplaces can leverage RFF as a seller-driven quality signal.
- Similar reward mechanisms can be implemented in other digital platforms.

Conclusion: Signaling can improve market efficiency by helping buyers differentiate high-quality sellers.

Other Examples / Comments?

Discussion Questions

Digital Advertising as Signaling:
Is (digital) advertising signaling? e.g., some brands that are already the first organic result for free pay for their own brand name keywords.

AI and Information Asymmetry:
Does AI **reduce** information frictions (better screening/matching), or **increase** them (cheaper fraud, fake reviews, synthetic identities)?